

Evaluating debiasing in lightweight BERT models

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Github repo: https://github.com/Ale-Dili/Progetto_NLP.

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Abstract

This paper investigates bias evaluation and mitigation in lightweight BERT models, using three variants as example. Three bias metrics has been used to assess stereotypical associations at different levels of language understanding. In order to reduce bias, Counterfactual Data Augmentation (CDA) has been adopted, which systematically swaps bias-sensitive attributes to create balanced training data. Results demonstrate that CDA effectively reduces stereotypical preferences across all models, bringing bias scores closer to the ideal 0.5 threshold. However, this bias reduction comes at a significant performance cost, particularly in text classification tasks where all debiased models show substantial capability degradation. The findings highlight the fundamental trade-off between bias mitigation and model performance, while demonstrating that simple, lightweight debiasing techniques can effectively address inherent biases in pre-trained language models. This work provides insights into practical bias mitigation strategies for deployment in sensitive applications.

Warning: *This report contains explicit statements of offensive stereotypes. These examples are included solely for the purpose of analyzing and measuring bias in language models, and do not reflect the views or opinions of the author.*

1 Introduction

Given the widespread use of language models, addressing bias within them has become crucial. Bias in language models refers to skewed representation toward specific perspectives, stereotypes, or demographic groups, which can negatively affect fairness in automated systems and reinforce existing prejudices. Since these models

are trained on large web-based text corpora, they inevitably reflect biases present in source data, making it essential to develop tools for identifying, measuring, and mitigating these biases in sensitive applications like recruitment, education, and healthcare. This project explores bias in lightweight, non-fine-tuned BERT models. BERT (Bidirectional Encoder Representations from Transformers) revolutionized NLP by understanding language through bidirectional context analysis, learning deep representations from large corpora like Wikipedia. While BERT is typically fine-tuned for specific applications, this study evaluates bias in its pre-trained form to assess inherent stereotypes before task-specific adaptation. The evaluation employs three different measures focusing on distinct levels of language understanding, followed by debiasing techniques and re-evaluation. To address potential performance degradation after debiasing, additional performance measurements are included to monitor any model capability loss.

2 Research question

The main goal of this project is to reduce bias toward specific categories in non-fine tuned BERT models. In addition to that, other question to be answered are:

- Which BERT model performs better?
- How much the fine tuning procedure can reduce bias?
- How to tackle the lack of structured data?

All of this, wrapped in a robust and modular framework to easily evaluate model biases and performance, and ready to support other BERT models.

3 Methodology

This chapter outlines the systematic approach used to evaluate and mitigate bias in non-fine-tuned BERT models. The methodology consists of three main components: bias measurement, a debiasing solution, and performance evaluation to assess potential trade-offs between bias reduction and model capability. As shown in Fig. 1, the pipeline of the project consists in evaluating BERT-based models on five different metrics (three bias related, and two performance related); applying Counterfactual Data Augmentation (CDA) as debiased technique, evaluating the debiased models, and then compare the result. The entire code was implemented to be as flexible as possible, allowing for an easy extension to new models.

3.1 Models

In this project, non-fine tuned BERT-based models were used. The model uses special tokens to structure inputs: [CLS] is prepended to sequences and its representation is used for classification tasks, [SEP] separates different text segments, and [MASK] tokens teach the model to predict missing words during pre-training. During inference, [CLS] provides sentence-level representations, [SEP] helps process multi-segment inputs, and the bidirectional attention learned through masking enables contextual

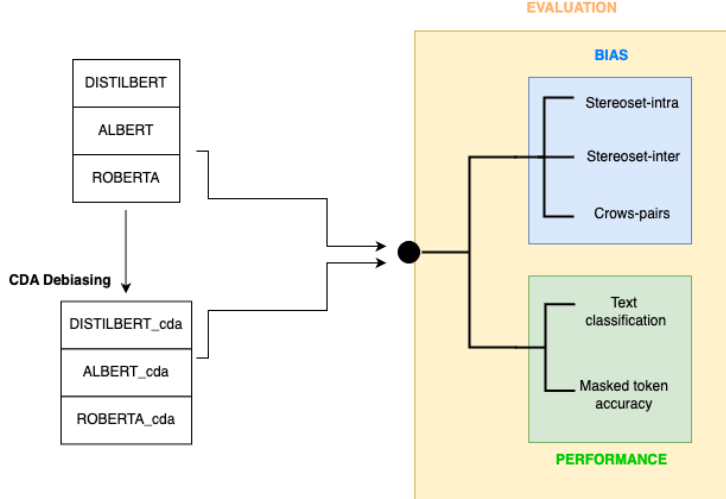


Fig. 1 General scheme of the framework

understanding of each token. Here three of BERT variants used in this project: DistilBERT, ALBERT, and RoBERTa.

DistilBERT

DistilBERT [1] is a compressed version of BERT that retains 97% of its language understanding while being 40% smaller and 60% faster. Using knowledge distillation during pre-training, it combines language modeling, distillation, and cosine-distance losses to maintain performance while reducing computational requirements.

ALBERT

ALBERT [2] addresses BERT’s scalability challenges through two parameter-reduction techniques: factorizing embedding matrices and sharing parameters across transformer layers. These optimizations significantly reduce memory consumption and training time while maintaining competitive performance, also incorporating sentence-order prediction for better inter-sentence modeling.

RoBERTa

RoBERTa [3] improves BERT through optimized training methodology rather than architectural changes. Key improvements include removing next-sentence prediction, dynamic masking, larger batch sizes, and byte-level BPE tokenization. These training optimizations achieve state-of-the-art results, demonstrating that pretraining procedures can be as important as model architecture.

3.2 Bias evaluation

Bias evaluation in language models requires specialized benchmarks that assess stereotypical associations across different levels of language understanding. The general idea is to count how many times the model prefers a stereotyped option, rather than a non-stereotyped one. An unbiased model should score 50% on each evaluation metric, where a score above this indicates a stereotyped model, and a score below indicates an antistereotyped model. To assess this preference, as [4] suggests, the Pseudo log-likelihood (PLL) measure was used. PLL is a method for evaluating the likelihood of a sequence commonly used with masked language models like BERT. Instead of computing the full joint probability of a sentence, PLL estimates it by summing the log probabilities of each token when predicted individually while conditioning on all other tokens in the sequence, providing an approximation that’s more efficient to calculate. The three metrics used to evaluate bias are: CrowS-Pairs, Stereoset intra-sentence, Stereoset inter-sentence

CrowS-Pairs

CrowS-Pairs [5] evaluates social bias at a token level by presenting a model two sentences: one that is more stereotyping and another that is less stereotyping.

Example 3.1. For example:

***Women** don’t know how to drive in the rain.*
***Men** know how to drive in the rain.*

The evaluation procedure involves iterating over each sentence and masking a single token at a time, measuring its log likelihood, and accumulating the result in a sum, while never masking the modified tokens that differ between the stereotypical and non-stereotypical sentences, as shown in Fig. 2. For each sentence the PLL is calculated and then the percentage of stereotyped preferences is computed.

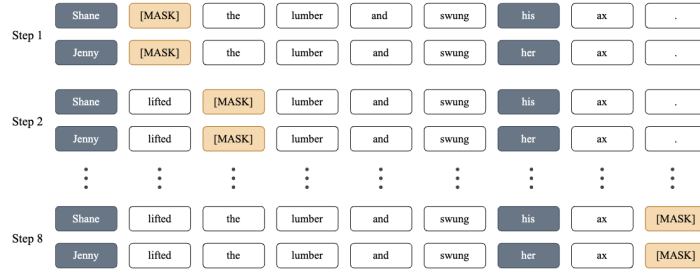


Fig. 2 CrowS-Pairs evaluation by iteratively masking the non-stereotyped word. *source:* [5]

Stereoset intra-sentence

StereoSet’s intrasentence task [6] evaluates bias through a fill-in-the-blank format that measures stereotypical associations. We can consider this evaluation at a sentence

level understanding. The evaluation determines which attribute the model assigns the highest likelihood, revealing whether it prefers stereotypical associations over counter-stereotypical ones. For this project the "unrelated" token has been ignored. In case of multiple tokens, the PLL is the average PLL of the tokens, as suggested in [4]. As for CrowS, a percentage of stereotypical preferences is then computed.

Choose the appropriate word:

Domain: Gender **Target:** Girl

Context: Girls tend to be more ____ than boys

Option 1: soft (stereotype)

Option 2: determined (anti-stereotype)

Option 3: fish (unrelated)

Fig. 3 Stereoset intra-sentence test. *source:* [6]

Stereoset inter-sentence

StereoSet’s intersentence task [6] evaluates bias through discourse-level reasoning by measuring sentence continuation preferences. The technique adopted for this evaluations consists in tokenizing the input composed as:

⟨Context⟩ [SEP] ⟨Stereo/Anti-stereo Sentence⟩.

Then an iteratively masking procedure on each token of the sentence is performed. The average PLL is taken and the sentence with the highest PLL is considered to be the choice of the model. We can summarize this with:

$$\arg \max_{sentence} \frac{1}{N} \sum_{i=0}^N \log(p(t_i)) \quad (1)$$

where N is number of token in a sentence, and t_i is the masked token at position i . The percentage of stereotyped preferences is finally computed.

Choose the appropriate sentence:

Domain: Race **Target:** Arab

Context: He is an Arab from the Middle East.

Option 1: He is probably a terrorist with bombs. (stereotype)

Option 2: He is a pacifist. (anti-stereotype)

Option 3: My dog wants a walk. (unrelated)

Fig. 4 Stereoset intra-sentence test. *source:* [6]

3.3 Performance evaluation

In order to evaluate any downgrade in performance of a model after the debiasing, two metrics were computed: Masked Token Accuracy and text classification

Masked Token Accuracy

Masked Token Accuracy (MTA) was performed on Open-web-text-10k [7], and it consists in masking random token (15% in this case) and then infer the model to take the word with the highest probability for the masked position. In addition to accuracy, top-k accuracy (with $k = 5$) is also performed.

Text classification

In order to evaluate a different capability of the model, text classification on [8] was performed. This dataset contains news labeled in four different classes. The idea behind the use text classification as a performance metric is related to a level of language understanding of the model. The setting evaluation was done by tweaking the prompt like in the following example:

(Original sentence) [SEP] The news topic is: [MASK].

and then the class with the higher probability in the masked position is taken. Precision, recall, f1 and accuracy were performed on a macro level.

3.4 Debiasing

As a technique to mitigate bias, Counterfactual Data Augmentation (CDA) [9] was implemented. CDA systematically swaps bias-sensitive attribute words in training datasets to create balanced corpora and break associations between gendered and gender-neutral words. CDA works by identifying sensitive attributes (such as gender pronouns, names, or occupation terms) and generating alternative versions of training examples with these attributes flipped, forcing models to learn task-relevant patterns rather than relying on biased correlations.

Example 3.2. For example a sentence like:

"He is a lawyer."

is flipped into

"She is a lawyer.";

in case of gender debias.

Implementation involves creating predefined word lists for sensitive attributes, applying automated substitution rules to generate counterfactual examples, and training models on the augmented balanced dataset. This approach is particularly effective because it directly addresses the root cause of bias in training data while preserving the original task structure and meaning.

	CrowS	Stereo-intra	Stereo-inter
DistilBERT	0.52	0.60	0.47
ALBERT	0.50	0.59	0.48
RoBERTa	0.49	0.53	0.46
DistilBERT_dca	0.55	0.52	0.46
ALBERT_dca	0.50	0.51	0.49
RoBERTa_dca	0.51	0.50	0.47

Fig. 5 Table of bias metrics

3.4.1 Using AI to fix AI

As with most machine learning problems, the main concern is the data. The absence of data for a specific task, lack of information about its structure, or poor data quality can all interfere effective training. A small experiment in this project explored whether a more powerful model could generate an artificial dataset to help debias a lightweight model. In this scenario, some typical issues were mitigated because the dataset generation followed strict guidelines. To perform Counterfactual Data Augmentation (CDA), we only need a sentence and then flip a stereotypical component to its opposite. GPT-4.1 Mini was used for this task. After a brief discussion about CDA, it was prompted to create a dataset that followed the necessary criteria to help mitigate biases related to race, gender, sexual orientation, and socioeconomic status. This generated dataset was then used to fine-tune the BERT models.

4 Results

The experimental evaluation reveals significant insights into both bias mitigation effectiveness and performance trade-offs across the three BERT-based models.

4.1 Bias evaluation

As shown in Fig. 5 applying DCA as a debias technique reduces the stereotyped decisions of the models. We can observe that DistilBERT is the less sensible model while ALBERT shows better performance improvement. Something that emerges from this numbers, is also the fact that even antistereotypical decisions have been mitigated, making the models more unbiased.

4.2 Performance evaluation

As shown in Fig. 6 we can observe that the bias reduction costs a lot in terms of model performance. We can see that RoBERTa, being the biggest model is less sensible and behave more predictably, both in MTA and Text Classification. One thing that is worth to mention is that MTA seems less robust for this evaluation, specifically for

	Masked token accuracy		Text Classification			
	Accuracy	Top 5 accuracy	Accuracy	Precision	Recall	F1
DistilBERT	0.47	0.80	0.51	0.66	0.52	0.44
ALBERT	0.18	0.43	0.64	0.71	0.65	0.62
RoBERTa	0.68	0.87	0.61	0.68	0.62	0.56
DistilBERT_dca	0.52	0.73	0.20	0.30	0.25	0.08
ALBERT_dca	0.39	0.63	0.19	0.04	0.25	0.08
RoBERTa_dca	0.58	0.80	0.20	0.30	0.25	0.09

Fig. 6 Table of performance metrics

lightweight models. While text classification shows how all the models completely lost their ability to classify text after the debiasing.

5 Conclusions

This report explored bias in lightweight BERT models, examining evaluation methods, reduction techniques, and performance trade-offs. The findings demonstrate that Counterfactual Data Augmentation (CDA) effectively reduces bias using a simple, lightweight approach requiring no specialized datasets. This is particularly valuable for fine-tuning applications, as incorporating CDA can help mitigate inherent biases in pre-trained representations.

This analysis reveals that robust models like RoBERTa show greater resilience to performance degradation during debiasing, while standard bias evaluation metrics may require deeper investigation to fully capture model behavior. The results highlight the fundamental tension between bias mitigation and task performance—CDA successfully reduces stereotypical associations but at a measurable cost to downstream capabilities.

Future research should investigate bias-performance trade-offs in fine-tuned contexts, specifically examining whether applying CDA to training datasets during task-specific fine-tuning can achieve effective bias reduction while maintaining task performance.

Declarations

AI Usage Disclosure: AI was used to support the development of this project, the drafting of descriptive texts, the writing of this document, and the identification of relevant datasets and references. The content has been reviewed and edited by a human. For more information on the extent and nature of AI usage, please contact the author.

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